

PRACTICE EXAM 3 OF 4 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 3

Twenty questions. More choose-two items and more judgement calls.

QUESTIONS 20 WEIGHTING 30/26/24/20 PASS MARK 14 of 20

An external audit firm with its own AWS account needs read access to one S3 bucket for a six-week engagement. The auditors already have IAM identities in their account.

Access must end automatically and the company will not create or manage identities for them.

Which solution is MOST appropriate?

- ☐ A Create IAM users for each auditor and issue access keys
- ☐ B Create a role in the company account trusting the auditors' account
- ☐ C Enable public read access on the bucket for the engagement
- ☐ D Generate presigned URLs for every object each morning

◆ PAUSE AND DECIDE

Which solution is MOST appropriate?

- A

 IAM users per auditor
Exactly what the question excludes, and long-lived keys leave the organisation.
- C

 Public bucket
Exposes audit data to the entire internet. Never correct.
- B

 Cross-account role
Auditors keep their own identities, credentials are temporary, and deleting the role ends access immediately.
- D

 Daily presigned URLs
Works for a handful of objects but does not scale to a bucket, and generating them daily is ongoing manual work.

THE SIGNAL

External party with their own AWS account plus temporary access means a cross-account role.

An existing unencrypted Amazon RDS for MySQL production database must be encrypted at rest using a customer managed key so that key usage can be audited.

The team asks whether encryption can be enabled during the next maintenance window.

What should the solutions architect tell them?

- ☐ A Enable encryption on the existing instance in a maintenance window
- ☐ B Snapshot the instance, copy the snapshot with encryption, and restore it
- ☐ C Create an encrypted read replica and promote it
- ☐ D Enable encryption on the underlying EBS volumes

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

- A

 Enable in a maintenance window
Not possible. Encryption cannot be turned on for an existing unencrypted RDS instance under any circumstances.
- C

 Encrypted replica then promote
A read replica of an unencrypted source cannot be encrypted, so the starting point makes this impossible.
- B

 Snapshot, encrypted copy, restore
The documented path. Encryption is applied during the snapshot copy and the restored instance is encrypted.
- D

 Encrypt the EBS volumes
RDS storage is managed. You have no access to the underlying volumes.

THE SIGNAL

Encrypting anything that already exists follows snapshot, encrypted copy, restore. Memorise it.

A company must prevent any account in its organisation from creating resources outside two approved Regions, and must also be alerted if an existing resource is found outside them.

Several accounts have their own administrators.

Which combination meets both requirements? (Choose two.)

- ☐ A A service control policy denying actions outside the approved Regions
- ☐ B An AWS Config rule detecting resources outside the approved Regions
- ☐ C IAM policies on every user and role denying other Regions
- ☐ D A CloudWatch alarm on the account's billing metric

◆ PAUSE AND DECIDE

Which combination meets both requirements? (Choose two.)

- A

 Service control policy
Preventive and applied above account administrators, satisfying the requirement that resources cannot be created.
- B

 Config rule for detection
Detective, which is exactly what the second requirement asks for: alerting on existing resources found outside the Regions.
- C

 IAM policies everywhere
A local administrator can detach them, and they must be maintained across every principal in every account.
- D

 Billing alarm
Alerts on spend, giving no indication of which Region a resource sits in.

THE SIGNAL

This question deliberately asks for one preventive and one detective control. Reading it as an either/or loses the mark.

Database credentials for an Amazon RDS instance must rotate every thirty days with no downtime and no manual steps. The team has no capacity to write or maintain rotation code.

The credentials are currently in an environment variable.

Which solution meets these requirements?

- ☐ A Secrets Manager with managed rotation for RDS
- ☐ B Parameter Store SecureString with a scheduled Lambda rotation function
- ☐ C An encrypted S3 object read at application start-up
- ☐ D A KMS-encrypted environment variable redeployed monthly

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A

 Secrets Manager managed rotation
Rotation for RDS is built in. It changes the database password and updates the stored secret together, with no code to maintain.
- B

 Parameter Store plus Lambda
Achieves rotation but requires writing and owning the function, which the question excludes.
- C

 Encrypted S3 object
Provides storage with no rotation, and reading at start-up means a restart to pick up changes.
- D

 Monthly redeployment
A manual step causing exactly the downtime the requirement forbids.

THE SIGNAL

'Rotation' means Secrets Manager. 'Cheap configuration values' would mean Parameter Store.

A company's public website suffered a two-hour outage when its TLS certificate expired. The certificate was bought externally, installed by hand, and the renewal reminder went to an employee who had left.

The company wants this to be impossible in future at the lowest cost. The site is served through CloudFront.

What should the solutions architect recommend?

- ☐ A Buy a three-year certificate and add a shared calendar reminder
- ☐ B Request a public certificate from AWS Certificate Manager
- ☐ C Use AWS Private Certificate Authority to issue the certificate
- ☐ D Store the certificate in Secrets Manager with rotation enabled

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Longer certificate and a reminder
Still depends on a human noticing. It postpones the same failure and costs money.
- B

 ACM public certificate
Free with CloudFront and renewed automatically, which removes the human dependency that caused the outage.
- C

 Private CA
Issues certificates trusted only inside the organisation. Public browsers would reject them, and it carries a monthly charge.
- D

 Secrets Manager
Secure storage does not renew a certificate.

THE SIGNAL

Public website plus lowest cost means ACM. Private CA is for internal trust and is never the cheap option.

Eighteen VPCs across two Regions are connected by a mesh of twenty-one peering connections. Adding a nineteenth VPC would require many new connections, and some teams report that certain VPC pairs cannot reach each other.

The networking team spends most of its time maintaining route tables.

Which solution reduces this complexity?

- ☐ A Complete the peering mesh with the missing connections
- ☐ B Replace the mesh with AWS Transit Gateway, peered across Regions
- ☐ C Consolidate all workloads into one VPC per Region
- ☐ D Replace peering with PrivateLink between every VPC pair

◆ PAUSE AND DECIDE

Which solution reduces this complexity?

- A

 Complete the mesh
Connections grow with the square of the VPC count. This makes maintenance worse, not better.
- B

 Transit Gateway
One hub replaces the mesh, routing is centralised, and a new VPC needs a single attachment. Peering extends it across Regions.
- C

 One VPC per Region
A vast migration that discards the workload isolation separate VPCs provide.
- D

 PrivateLink everywhere
PrivateLink exposes a specific service endpoint, not general routing between networks.

THE SIGNAL

'Some VPC pairs cannot reach each other' is peering's non-transitivity described without being named.

A stateless web tier runs across three Availability Zones in an Auto Scaling group. The team proposes an active-active second Region at roughly double the cost, to improve availability. The application has had three outages in two years, all caused by faulty deployments rather than infrastructure failure.

What should the solutions architect advise?

- ☐ A Proceed, since multi-Region is the highest-availability design
- ☐ B Advise that a second Region would not have prevented any of the outages, and improve the deployment process
- ☐ C Add a fourth Availability Zone instead
- ☐ D Replace Auto Scaling with reserved capacity

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 Proceed with multi-Region
Doubles cost against a failure mode that has never occurred, while leaving the actual cause untouched.
- B

 Fix the deployment process
A bad deployment propagates to every Region, so multi-Region would have produced two broken Regions. Match the remedy to the failure.
- C

 Fourth AZ
Three zones already tolerate a zone failure. A fourth spends money on a covered risk.
- D

 Reserved capacity
Addresses capacity availability, which caused none of the outages.

THE SIGNAL

Read the failure history. The exam does include 'do not build the impressive thing' answers.

An internal reporting system is busy between 07:00 and 09:00 each weekday and idle otherwise. Instances take about eight minutes to boot and warm caches.

With target tracking on CPU, users report the first half hour each morning is slow.

What is the MOST appropriate change?

- ☐ A Lower the target tracking CPU threshold
- ☐ B Add scheduled scaling to raise capacity before 07:00
- ☐ C Run at maximum capacity continuously
- ☐ D Replace the group with one larger instance

◆ PAUSE AND DECIDE

What is the MOST appropriate change?

- A

 Lower the threshold
Still reactive, so with eight-minute boots capacity still arrives after the load, and it raises cost all day.
- C

 Maximum always
Pays for peak capacity twenty-four hours a day to serve a two-hour workload.
- B

 Scheduled scaling
The load is predictable by the clock, so capacity can be in place before demand rather than chasing it.
- D

 One larger instance
Removes redundancy and still must be sized for peak.

THE SIGNAL

Predictable by the clock plus slow start-up means scheduled scaling. Reactive policies always arrive late.

A global SaaS product must stay available if an entire Region fails, tolerate only seconds of data loss, and require no manual intervention. Users are in Europe, North America and Asia, with latency complaints from Asia.

The data store is DynamoDB and budget for multi-Region is approved.

Which approach best meets these requirements?

- ☐ A DynamoDB global tables with the application deployed in three Regions
- ☐ B DynamoDB with point-in-time recovery and nightly cross-Region backups
- ☐ C DynamoDB in one Region with DAX clusters in the others
- ☐ D DynamoDB with a read replica in each Region

◆ PAUSE AND DECIDE

Which approach best meets these requirements?

- A

 Global tables in three Regions
Multi-active replication in seconds, local reads and writes for every user, and a Region failure leaves the others serving.
- B

 PITR and nightly backups
Nightly backups cannot meet a seconds-level RPO, and restoring is manual intervention.
- C

 DAX in other Regions
DAX is an in-Region read cache holding no durable copy. A Region failure loses the service.
- D

 Read replicas per Region
DynamoDB has no such feature. Cross-Region replication is provided by global tables.

THE SIGNAL

This is a rare case where multi-Region is genuinely correct, because the question demands every element of it.

A disaster recovery plan claims a two-hour RTO based on restoring snapshots into a second Region and running CloudFormation templates. It has never been tested.

A tabletop exercise found the templates reference an AMI that exists only in the primary Region, and nobody knows how long a restore takes.

What should the solutions architect recommend FIRST?

- ☐ A Upgrade to warm standby to reduce reliance on the restore
- ☐ B Run a full DR test to measure actual RTO and fix what it reveals
- ☐ C Copy the AMIs to the second Region and consider the plan complete
- ☐ D Increase snapshot frequency to improve RPO

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

- A

 Upgrade to warm standby
Spends money to avoid a process nobody has measured. The current strategy may well meet two hours once fixed.
- B

 Test, measure, fix
An untested plan has an unknown RTO. Testing validates the objective and surfaces the errors the exercise has already hinted at.
- C

 Copy AMIs and stop
Fixes the one fault found by accident, strongly suggesting others remain undiscovered.
- D

 More frequent snapshots
Improves a metric nobody raised. The stated concern is recovery time.

THE SIGNAL

An untested DR plan does not have an RTO, it has an aspiration.

A company migrating a legacy system to AWS over six months must run both the old and new systems live, shifting traffic gradually in ten percent increments.

They need to monitor error rates after each shift and revert instantly if problems appear.

Which Route 53 routing policy supports this?

- ☐ A Failover routing
- ☐ B Weighted routing
- ☐ C Latency-based routing
- ☐ D Multivalue answer routing

◆ PAUSE AND DECIDE

Which Route 53 routing policy supports this?

A Failover

Active-passive. All traffic goes to the primary until it breaks. It cannot split by proportion.

C Latency-based

Routes by which endpoint is fastest for each user, not by a proportion you control.

B Weighted

Assigns a proportion to each endpoint, so ten percent increments and instant reversion are simply weight changes.

D Multivalue answer

Returns several healthy records for basic distribution with no control over the split.

THE SIGNAL

Proportion means weighted. Performance means latency. Location means geolocation. Availability means failover.

A company runs twenty microservices in containers on self-managed EC2 hosts. The platform team spends most of its time patching hosts, planning cluster capacity and handling scaling.

Traffic is steady and modest. The company does not use Kubernetes.

Which solution meets this with the LEAST operational overhead?

- | | |
|--|--|
| ☐ A Amazon EKS with managed node groups | ☐ B Amazon ECS with the Fargate launch type |
| ☐ C Amazon ECS on EC2 with Auto Scaling | ☐ D Lambda container images |

◆ PAUSE AND DECIDE

Which solution meets this with the LEAST operational overhead?

- A

EKS with managed nodes
Introduces Kubernetes to a team that does not use it, adding a control plane charge and substantial new complexity.
- B

ECS with Fargate
No hosts to patch, no capacity to plan, and containers run unchanged. Removes all three of the team's stated tasks.
- C

ECS on EC2 with Auto Scaling
Automates scaling but leaves patching and capacity planning, which is most of the current workload.
- D

Lambda container images
Lambda runs container images, but converting twenty long-running services to event-driven functions is a rewrite.

THE SIGNAL

Containers plus LEAST overhead plus no Kubernetes mention equals ECS on Fargate.

A read-heavy product catalogue on Amazon RDS is at eighty-five percent CPU during business hours. Ninety-five percent of queries request the same three thousand popular items.

The application team can change the code and wants to reduce database load.

What is the MOST effective change?

- ☐ A Add Amazon ElastiCache and cache the popular queries
- ☐ B Add a DynamoDB Accelerator cluster
- ☐ C Enable Multi-AZ so the standby serves reads
- ☐ D Migrate the catalogue to Amazon Redshift

◆ PAUSE AND DECIDE

What is the MOST effective change?

- A

ElastiCache
A small hot subset serving most requests is the ideal caching pattern. Repeat reads never reach the database.
- B

DAX
DAX only works with DynamoDB. This is an RDS database, so it cannot be placed in front of it.
- C

Multi-AZ for reads
The Multi-AZ standby serves no traffic at all. This is the classic confusion with read replicas.
- D

Redshift
A warehouse for analytical queries, not high-volume lookups of individual products.

THE SIGNAL

Ninety-five percent of traffic on a small subset is the textbook cache signature. DAX is only ever for DynamoDB.

A retailer ingests point-of-sale events. A fraud engine must react within two seconds, an inventory service consumes the same events, and an analytics team periodically reprocesses the last seven days after changing its logic.

No consumer may miss an event.

Which service should carry the events?

- ☐ A Kinesis Data Firehose delivering to S3
- ☐ B Kinesis Data Streams
- ☐ C A single SQS standard queue
- ☐ D SNS with email subscriptions

◆ PAUSE AND DECIDE

Which service should carry the events?

- A

Firehose
Buffers for at least sixty seconds, breaking the two-second requirement, and it delivers rather than serving live consumers.
- B

Data Streams
Multiple consumers read the same records independently, latency is sub-second, and retention allows replay of the last seven days.
- C

Single SQS queue
A message is consumed once, so the first reader denies it to the others.
- D

SNS to email
No storage means no replay, and email is not an ingestion mechanism for high-volume events.

THE SIGNAL

Replay is the sharpest discriminator. Only Data Streams retains records for consumers to re-read.

A video rendering workload writes large temporary files during each job and reads them back within the same job. The files are discarded when the job ends. The chosen instance type includes local NVMe storage.

The team currently writes to Provisioned IOPS EBS and finds it slow and expensive.

What should the solutions architect recommend?

- ☐ A Write temporary files to the instance store volumes
- ☐ B Increase the provisioned IOPS on the EBS volume
- ☐ C Write temporary files to Amazon EFS
- ☐ D Write temporary files to S3 with Transfer Acceleration

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Instance store
Physically attached NVMe gives the highest IOPS available, costs nothing extra, and its volatility is irrelevant for discarded files.
- B

 More provisioned IOPS
Raises both performance and cost when local storage already included in the instance price is faster.
- C

 EFS
A network file system is slower than local disk and adds cost. Nothing here needs sharing.
- D

 S3 with Transfer Acceleration
Object storage for scratch files, and Transfer Acceleration addresses long-distance uploads, not local performance.

THE SIGNAL

Instance store's weakness is irrelevant when the data is discarded anyway. That is the whole question.

A four terabyte Oracle database must move to Amazon Aurora PostgreSQL. The application must keep running, with a cutover window of fifteen minutes.

The schema uses Oracle-specific data types and stored procedures with no direct PostgreSQL equivalent.

Which combination should be used?

- ☐ A DMS with the Schema Conversion Tool and ongoing replication
- ☐ B Export via Snowball and import into Aurora
- ☐ C Take an Oracle dump, restore into Aurora, then switch over
- ☐ D Use DataSync to copy the database files

◆ PAUSE AND DECIDE

Which combination should be used?

- A

 DMS with SCT and replication
SCT converts the incompatible schema and procedures; DMS does a full load then replicates changes, so cutover is only the final catch-up.
- B

 Snowball
Addresses bandwidth, which is not the problem at four terabytes, and provides no continuous replication.
- C

 Dump and restore
Requires downtime for the whole load, and a native dump does not convert Oracle types to PostgreSQL.
- D

 DataSync
Moves files between file and object storage. You cannot write database files into Aurora's managed storage.

THE SIGNAL

Different engines means SCT. Minimal downtime means DMS ongoing replication.

An API receives around six hundred requests per day, arriving unpredictably. It runs on two continuously running instances behind a load balancer. Each request completes in under three seconds.

The team wants lower cost and is willing to change the deployment model.

Which architecture is MOST cost-effective?

- ☐ A Reserved Instances for the two existing instances
- ☐ B API Gateway with AWS Lambda
- ☐ C One smaller instance without the load balancer
- ☐ D Fargate tasks running continuously behind the load balancer

◆ PAUSE AND DECIDE

Which architecture is MOST cost-effective?

A Reserve the instances

Discounts the rate while still paying for two instances and a load balancer running all day for six hundred short requests.

C One smaller instance

Halves the cost and removes redundancy while still paying continuously for an almost always idle workload.

B API Gateway with Lambda

Six hundred short invocations a day costs very little, nothing runs between requests, and the load balancer charge disappears.

D Fargate continuously

Removes host management but keeps paying for always-running tasks and the load balancer.

THE SIGNAL

Discounting idle capacity is always worse than eliminating it.

A company generates intermediate processing files in S3. A downstream job reads them within hours and they are never needed afterwards, but nobody deletes them. The bucket holds three hundred terabytes.

The files can be regenerated from source data at any time.

Which combination is MOST cost-effective? (Choose two.)

- ☐ A Store the files in S3 One Zone-IA
- ☐ B Add a lifecycle rule expiring the files after seven days
- ☐ C Transition the files to Glacier Deep Archive after one day
- ☐ D Enable versioning so old files can be recovered

◆ PAUSE AND DECIDE

Which combination is MOST cost-effective? (Choose two.)

- A

 One Zone-IA
Re-creatable data is exactly what One Zone-IA is for, and it costs about twenty percent less than Standard-IA.
- B

 Expire after seven days
The data is never needed after a few hours. Deleting removes the cost entirely rather than reducing it.
- C

 Deep Archive after a day
Archiving data never read again pays to keep it forever, plus a 180-day minimum charge.
- D

 Versioning
Stores additional copies of re-creatable data, increasing cost while solving nothing.

THE SIGNAL

The cheapest storage class is no storage. Expiry beats archiving when data is genuinely finished with.

A development team runs sixty EC2 instances for testing. Engineers work 09:00 to 18:00 on weekdays and the instances run continuously.

Management wants cost reduction without affecting engineers during business hours and without any long-term commitment.

Which approach delivers the LARGEST saving?

- ☐ A One-year Reserved Instances for the sixty instances
- ☐ B A scheduled stop and start outside working hours
- ☐ C Move the instances to Spot
- ☐ D Right-size the instances one size smaller

◆ PAUSE AND DECIDE

Which approach delivers the LARGEST saving?

- A

 Reserved Instances
Excluded by 'no long-term commitment', and it would discount hours nobody needs.
- B

 Scheduled stop and start
45 working hours out of 168 in a week, so stopping the rest removes roughly three quarters of the compute bill with no commitment.
- C

 Spot
Cheap per hour, but engineers losing environments mid-task affects their ability to work.
- D

 Right-size one size down
Saves perhaps half the cost of instances that are idle three quarters of the time anyway.

THE SIGNAL

Development environments running out of hours are the most common cost question on this exam.

A company running one hundred and eighty workloads across twenty-two accounts wants regular identification of idle load balancers, underutilised instances, unassociated Elastic IPs, and service limits being approached.

They will not build custom tooling, and they have Enterprise Support.

Which service should be used?

- | | |
|--|---|
| ☐ A AWS Trusted Advisor | ☐ B AWS Cost Explorer |
| ☐ C AWS Well-Architected Tool | ☐ D Amazon CloudWatch dashboards |

◆ PAUSE AND DECIDE

Which service should be used?

- A

Trusted Advisor
Its checks cover this exact list including service limits, with nothing to build, and Enterprise Support unlocks the full set.
- B

Cost Explorer
Shows spend by service and account. It does not identify specific idle resources or service limits.
- C

Well-Architected Tool
A questionnaire-based review framework. It does not inspect resources automatically.
- D

CloudWatch dashboards
Displays metrics you configure, which is the custom tooling the question rules out.

THE SIGNAL

Service limits appear in no other option. That single item identifies Trusted Advisor.

Scoring Practice Exam 3

By now the trend across three papers matters more than any single score.

- ◆ **15 or above and rising across all three.** You are ready. Use Exam 4 as a final check close to your exam date.
- ◆ **Consistent 14 to 16 with no trend.** Solid but not comfortable. Target the specific question types you keep missing.
- ◆ **Choose-two questions missed repeatedly.** You are answering half. Reread every stem for the number of answers required.
- ◆ **Judgement questions missed repeatedly.** Review modules 5, 6 and 14. The impressive answer is often wrong.

ON THE EXAM

Three papers is enough data for a trend. A flat line means changing how you revise, not revising more.